

The L^2 -Index Theorem for Homogeneous Spaces of Lie Groups

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Overview and Purpose

This paper proves that for a G -invariant elliptic pseudodifferential operator P on a homogeneous space $M = G/H$, the *analytical index* $\text{ind}_G P$ —defined via a canonical von Neumann trace on the commutant of the left regular representation of G —equals the *topological index* $\text{ind}_t[\sigma(P)]$ —defined via characteristic classes in the relative Lie algebra cohomology of the pair (G, H) . This is the Atiyah-Singer theorem stripped of compactness: the manifold M is non-compact (it is the homogeneous space of a generally non-compact Lie group G), and ordinary Fredholm theory is unavailable. The paper also shows that $\ker P$, as a representation of G , is a finite direct sum of irreducible discrete series representations, giving both existence and vanishing criteria for L^2 -solutions to G -invariant elliptic equations.^{[1][2]}

Two results are proved: (1) the index formula $\text{ind}_G P = \text{ind}_t[\sigma(P)]$ in full generality for unimodular G ; and (2) for a large class of groups, the representation of G on $\ker P$ is a finite direct sum of irreducible square-integrable (discrete series) representations.^[^1]

Background: The Atiyah-Singer Theorem and Its Covering Generalization

The Classical Compact Case

The Atiyah-Singer index theorem concerns an elliptic differential operator $D : C^\infty(M, E) \rightarrow C^\infty(M, F)$ on a *compact* manifold M . Because M is compact, D is Fredholm (finite-dimensional kernel and cokernel), and its *analytical index*^[^3]

$$\text{ind } D = \dim \ker D - \dim \ker D^*$$

is an integer. The theorem states that this equals a *topological index* built from the principal symbol $\sigma(D)$ via characteristic classes — the Chern character and the $\hat{A}$ -genus. The result is famous because it identifies a purely analytic datum (the dimension of a solution space) with topology that can be computed from the symbol alone, without solving any equations.^{[4][5]}

The key examples are: the de Rham operator $(d + d^*)$, whose index is the Euler characteristic; the Dolbeault operator, whose index is the holomorphic Euler characteristic (Riemann–Roch); and the Dirac operator, whose index is the $\hat{A}$ -genus.

The L^2 -Index Theorem of Atiyah for Covering Spaces

When M is *non-compact*, ordinary Fredholm theory breaks down. The kernel and cokernel of D can be infinite-dimensional, so the naive index $\dim \ker D - \dim \ker D^*$ is meaningless. Atiyah's 1976 theorem handles the special case where M is a Galois covering of a compact manifold: let Γ be a discrete group acting freely and properly discontinuously on M with compact quotient $\bar{M} = \Gamma \backslash M$. If D is Γ -invariant, one replaces ordinary dimension by the Γ -*dimension*: for a Γ -invariant closed subspace $\mathcal{H} \subset L^2(M, E)$, set^{[6][7][^8]}

$$\dim_{\Gamma} \mathcal{H} := \operatorname{tr}_{\Gamma} K_{\mathcal{H}},$$

where $K_{\mathcal{H}}$ is the orthogonal projection onto $\mathcal{H}$ and $\operatorname{tr}_{\Gamma}$ is the canonical semifinite trace on the commutant of the Γ -action (the von Neumann trace, assigning each translate a weight proportional to $1/|\Gamma|$ locally). The Γ -*index* is then $\operatorname{ind}_{\Gamma} D = \dim_{\Gamma} \ker D - \dim_{\Gamma} \ker D^*$, which is now a real number. Atiyah's theorem states $\operatorname{ind}_{\Gamma} D = \operatorname{ind} \bar{D}$, where $\bar{D}$ is the descended operator on the compact quotient. The significance: $\operatorname{ind}_{\Gamma} D > 0$ guarantees the existence of nontrivial L^2 -solutions on the non-compact cover^{[6][9]}.

This was used crucially by Atiyah-Schmid to construct discrete series representations of semisimple Lie groups as spaces of L^2 -harmonic spinors on G/K , where K is a maximal compact subgroup.^[^1]

The Gap Addressed by This Paper

Atiyah's theorem requires a *cocompact discrete subgroup* $\Gamma \subset G$. Outside the semisimple case, most unimodular Lie groups have no such subgroup, and even when Γ exists, the dependence of $\operatorname{ind}_{\Gamma} D / \operatorname{covol}(\Gamma)$ on Γ must be shown to vanish. Connes-Moscovici define the G -index *intrinsically*, without any covering construction, and prove the index formula for *all unimodular Lie groups*.^{[10][1]}

The Setup: Homogeneous Spaces and Invariant Operators

The Geometric Data

Let G be a unimodular Lie group (left and right Haar measures coincide) with at most countably many connected components, $H \subset G$ a compact subgroup, and $M = G/H$ the homogeneous space of left cosets $\{gH : g \in G\}$. The G -action on M is by left multiplication: $g_0 \cdot (gH) = g_0 g H$. The base point is $o = 1 \cdot H \in M$.^[^1]

Given a finite-dimensional unitary representation $\varepsilon : H \rightarrow U(E)$, the *homogeneous vector bundle* $\xi = G \times_H E$ over M is the quotient of $G \times E$ by the equivalence $(g, v) \sim (gh, \varepsilon(h)^{-1}v)$. Its smooth sections are identified with

$$C^{\infty}(M, \xi) \cong \{f \in C^{\infty}(G) \otimes E : f(gh) = \varepsilon(h)^{-1} f(g), \forall h \in H\} = (C^{\infty}(G) \otimes E)^H,$$

where H acts via the right regular representation $R(h)$ combined with $\varepsilon(h)$. Similarly the L^2 -sections are $L^2(M, \xi) \cong (L^2(G) \otimes E)^H$. The left action $L(g_0)f(g) = f(g_0^{-1}g)$ preserves this H -invariant subspace and defines a unitary representation of G on $L^2(M, \xi)$.^[^1]

G-Invariant Pseudodifferential Operators

Let ξ_1, ξ_2 be homogeneous bundles. A pseudodifferential operator (Ψ DO) $P \in \Psi^n(M, \xi_1, \xi_2)$ of order n is called *G-invariant* if $L(g)PL(g)^{-1} = P$ for all $g \in G$. For such an operator, the distributional kernel $K_P(x, y)$ satisfies $K_P(x, y) = k_P(x^{-1}y)$ for a distribution k_P on G . This is a convolution structure: $(Pu)(x) = \int_G k_P(x^{-1}y)u(y) dy$. The operator is fully determined by its kernel value at the identity element $1 \in G$.^[1]

A G -invariant Ψ DO P is said to be *G-compactly supported* (written $P \in \Psi_c^n(M, \xi_1, \xi_2)^G$) if k_P has compact support, i.e., $K_P(x, y) = 0$ when $x^{-1}y$ lies outside a compact set. For the theory to work on the non-compact space M , this compactness condition replaces the global compactness of the manifold in the classical setting. It ensures compositions of operators are well-defined, and that the G -trace (introduced in §2) is finite.^[1]

Section 1: The Averaging Operation and the Calculus

Constructing G-Invariant Operators by Averaging

Any ordinary compactly supported Ψ DO $P \in \Psi_{cc}^n(M, \xi_1, \xi_2)$ (with compact kernel support, but no G -invariance) can be *averaged* over G to produce a G -invariant one:

$$\text{Av}(P)u = \int_G L(g)PL(g)^{-1}u dg, \quad u \in C_c^\infty(M, \xi_1).$$

The integral is well-defined because the compact support of K_P means only a compact subset of G contributes. Lemma 1.1 states that $\text{Av}(P) \in \Psi_c^n(M, \xi_1, \xi_2)^G$ and its symbol is the average of the symbol: $\sigma(\text{Av}(P)) = \int_G L(g)\sigma(P)L(g)^{-1} dg$. Conversely, every G -invariant Ψ DO arises as an average.^[1]

A *cut-off function* is a smooth $f : G \rightarrow \mathbb{R}_{>0}$ such that: (a) $f(gh) = f(g)$ for $h \in H$ (right H -invariance, so f descends to M); (b) $\int_G f(x) dx = 1$. Lemma 1.2 states that for $P \in \Psi_c^n(M, \xi_1, \xi_2)^G$, one has $P = \text{Av}(fP)$ for any cut-off function. This is the mechanism by which global integrals over G are reduced to local traces: multiplying by f localizes the operator, and averaging restores G -invariance.^[1]

Using this, every elliptic G -invariant Ψ DO has a *G-invariant parametrix*: Proposition 1.3 states that if $P \in \Psi_c^n(M, \xi_1, \xi_2)^G$ is elliptic, then there exists $Q \in \Psi_c^{-n}(M, \xi_2, \xi_1)^G$ (a G -invariant approximate inverse) such that $I - QP \in \Psi_c^{-\infty}(M, \xi_1, \xi_1)^G$ and $I - PQ \in \Psi_c^{-\infty}(M, \xi_2, \xi_2)^G$. The proof: take an ordinary parametrix Q' , then set $Q = \text{Av}(fQ')$. This works because the error $S' = I - Q'P$ is smoothing, and averaging a smoothing operator gives a G -invariant smoothing operator.^[1]

Boundedness of Zero-Order G -Invariant Operators

In the compact setting, any zero-order Ψ DO is bounded on L^2 . On a non-compact homogeneous space, this is not automatic for an arbitrary Ψ DO, but Proposition 1.4 and Corollary 1.6 show it does hold for G -invariant operators of order zero: any $P \in \Psi^0(M, \xi_1, \xi_2)^G$ extends to a bounded operator $P : L^2(M, \xi_1) \rightarrow L^2(M, \xi_2)$. The proof uses Lemma 1.5, a Hilbert-space estimate: if $F \in L^2(G, \mathcal{K})$ and inner products $\langle F(x), F(y) \rangle = 0$ for $x^{-1}y \notin K$ (a compact set), then $\|\int F(x) dx\|^2 \leq |K| \int \|F(x)\|^2 dx$. Applied to $F(x) = L(x)PL(x)^{-1}u$, this turns the non-compactness of G into a finite bound [^1].

The C^* -Algebra Exact Sequence

This leads to the key short exact sequence (Proposition 1.7 and Corollary 1.8): for $\xi_1 = \xi_2 = \xi$,

$$0 \rightarrow C_G^*(\xi) \rightarrow \Psi_G^*(\xi) \xrightarrow{\sigma_0} C_H(S, E, E) \rightarrow 0$$

where: $C_G^*(\xi)$ is the norm-closure of the smoothing operators (the "ideal of compact-like operators" in this setting); $\Psi_G^*(\xi)$ is the norm-closure of zero-order G -invariant Ψ DOs; and $C_H(S, E, E)$ is the space of continuous H -equivariant functions from the unit sphere $S \subset V = (\mathfrak{g}/\mathfrak{h})^*$ to $\text{End}(E)$, carrying the symbol $\sigma_0(P)$ of P . The map σ_0 extends the principal symbol to the full C^* -algebra closure. The exactness says the symbol map is surjective (any smooth elliptic symbol is realizable) and its kernel is exactly the compacts (the obstruction to invertibility is entirely in the symbol).[^1]

Section 2: The Natural G -Trace

The von Neumann Trace on the Commutant

The commutant $\mathcal{M}_G$ of the left regular representation L of the unimodular Lie group G on $L^2(G)$ is the von Neumann algebra generated by the right regular representation R . It carries a canonical faithful normal semifinite trace tr_G , uniquely determined by

$$\text{tr}_G(R(f)^*R(f)) = \int_G |f(g)|^2 dg$$

for $f \in L^2(G)$ with $R(f)$ bounded. Here $(R(f)u)(x) = \int f(g)u(xg) dg$. This trace is the non-compact analogue of the "dimension per fundamental domain": on a compact group, tr_G would simply count dimensions weighted by the volume of the group.[^1]

For a homogeneous bundle ξ associated to a representation ε of H on E , the commutant $\mathcal{M}_G(\xi)$ of the G -action on $L^2(M, \xi) \cong (L^2(G) \otimes E)^H$ is the reduced algebra from $\mathcal{M}_G \otimes \text{End}(E)$ corresponding to the H -invariant subspace. Its trace tr_G combines the group trace with the ordinary matrix trace on $\text{End}(E)$. A key formula (Corollary 2.3) is:

$$\text{tr}_G(R(\varphi)) = \varphi(1), \quad \varphi \in (C_c^\infty(G) \otimes \text{End}(E))^{H \times H},$$

i.e., the G -trace of the convolution operator with kernel φ equals the value of φ at the identity. This is the bridge between the abstract von Neumann trace and concrete local expressions.[^1]

Connecting G-Trace to Ordinary Trace: The Operator-Valued Weight

The critical technical result of Section 2 is that the G-trace and the ordinary operator trace are related by the averaging operation. There is a canonical *operator-valued weight* $\text{Av}_G : \mathcal{M}_G^+(\xi) \rightarrow \mathcal{B}^+(\xi)$ defined by

$$\text{Av}_G(T) = \int_G L(g) T L(g)^{-1} dg,$$

with the property $\text{tr}_G(\text{Av}_G(T)) = \text{tr}(T)$ for any positive $T \in \mathcal{B}(\xi)$. Lemma 2.1 quantifies this: for self-adjoint T in the intersection of the domains, $\|\text{Av}_G(T)\|_{\text{tr}_G} \leq \|T\|_{\text{tr}}$ and $\text{tr}_G(\text{Av}_G(T)) = \text{tr}(T)$. Lemma 2.2 identifies the two averagings: for $P \in \Psi_{cc}^{-\infty}(\xi)$, one has $\text{Av}_G(P) = \text{Av}(P)$ as defined in Section 1. Together these mean: to compute a G-trace, one can instead compute an ordinary trace on a localized version of the operator, then use the cut-off function to normalize [^1].

In practice, this means: $\text{tr}_G(P) = \text{tr}(fP) = \int_G f(x) K_P(x, x) dx = k_P(1)$ for any zero-order G -invariant Ψ DO. The G-trace is just the kernel diagonal value at the identity, irrespective of the non-compactness of M . [^1]

Section 3: The Analytical Index

Definition

Given a G -invariant elliptic Ψ DO $P \in \Psi_c^n(\xi_1, \xi_2)^G$, define the *G-dimension* of a closed G -invariant subspace $\mathcal{W} \subset L^2(M, \xi)$ by

$$\dim_G \mathcal{W} := \text{tr}_G K_{\mathcal{W}},$$

where $K_{\mathcal{W}}$ is the orthogonal projection onto $\mathcal{W}$. Lemma 3.2 shows $\dim_G \ker P < \infty$: take a G -invariant parametrix Q (Proposition 1.3), so $S_1 = I - QP \in \Psi_c^{-\infty}(\xi_1)^G$. For any $u \in \ker P$, $u = S_1 u$, so $K_{\ker P} \leq S_1 K_{\ker P}$, and since S_1 is of G-trace class (by Corollary 2.3), $\dim_G \ker P \leq \text{tr}_G(S_1) < \infty$. [^1]

The *analytical index* of P is defined as

$$\text{ind}_G P := \dim_G \ker P - \dim_G \ker P^*.$$

This is a real number (not necessarily an integer), and it reduces to the ordinary Fredholm index when G is compact. The non-integrality for non-compact G is not a defect: it is a feature, as illustrated by the nilpotent example in Section 7.2 where $\text{ind}_G A_{p,q}^+ = |p + \theta q|$ with θ irrational [^1].

Parametrix Formula for the G-Index

Lemma 3.3 gives an equivalent expression: if Q is a G -invariant parametrix for P , so that $S_1 = I - QP$ and $S_2 = I - PQ$ are G -invariantly smoothing, then

$$\text{ind}_G P = \text{tr}_G S_1 - \text{tr}_G S_2.$$

This is the direct analogue of the classical formula $\text{ind } P = \text{tr}(I - QP) - \text{tr}(I - PQ)$ for Fredholm operators on compact manifolds. The proof runs the same formal algebra, with the G -trace replacing the ordinary trace, and uses the key fact that $\text{tr}_G(AB) = \text{tr}_G(BA)$ within the appropriate domains.[^1]

Properties of the G-Index

Proposition 3.4 establishes the standard index calculus:

- (i) **Additivity:** $\text{ind}_G(P_2 P_1) = \text{ind}_G P_1 + \text{ind}_G P_2$.
- (ii) **Homotopy invariance:** ind_G is locally constant on the space of elliptic operators in the norm topology. Two operators with symbols close in $C_H(S, E_1, E_2)$ have the same G -index.
- (iii) **Stability under lower-order perturbation:** For $P \in \Psi_c^0(\xi_1, \xi_2)^G$ elliptic and $S \in \Psi_c^{-\infty}(\xi_1, \xi_2)^G$, $\text{ind}_G(P + S) = \text{ind}_G P$.

The proof of (iii) uses the parametrix formula: adding S to P changes $S_1 = I - QP$ to $S_1 - QS$, and by the trace property $\text{tr}_G(QS) = \text{tr}_G(SQ)$, the change cancels in the difference $\text{tr}_G S_1 - \text{tr}_G S_2$.[^1]

Factorization Through the Symbol: The Index as a K-Theory Map

Proposition 3.5 is the key structural result: the G -index factors through the equivariant K-theory group $K_H(V)$, where $V = \mathfrak{g}^*|_{\mathfrak{h}^0} = \{t \in \mathfrak{g}^* : t|_{\mathfrak{h}} = 0\}$ is the *co-isotropy representation* of H . Specifically, there is a unique group homomorphism

$$\text{ind}_a : K_H(V) \rightarrow \mathbb{R}$$

such that $\text{ind}_a[\sigma_0(P)] = \text{ind}_G P$ for every G -invariant elliptic Ψ DO P .

To understand $K_H(V)$: it is the equivariant K-theory (with compact supports) of the real vector space V , regarded as an H -space via the coadjoint action. Concretely, elements of $K_H(V)$ are represented by smooth H -equivariant maps $\alpha : S \rightarrow \text{Iso}(E_1, E_2)$ (isomorphisms between H -modules, parameterized over the unit sphere $S \subset V$), up to homotopy and isomorphism. The symbol of an order-zero elliptic Ψ DO is exactly such a map.[^1]

Three checks are needed for well-definedness. (1) Isomorphic symbols give the same G -index: if $\alpha \cong \alpha'$ via bundle isomorphisms φ, ψ , then the corresponding operators differ by a lower-order term, and stability (3.4(iii)) applies. (2) Homotopic symbols give the same G -index: if α_0, α_1 are close, the operators P_0, P_1 satisfy $\|P_0 - P_1 + S\| < 2\delta$ for some smoothing S (by Corollary 1.8), so local constancy of ind_G applies. (3) Constant symbols give zero index: the corresponding operator is simply a bundle isomorphism, which is invertible, so $\ker P = \ker P^* = 0$ [^1].

Corollary 3.6: If $M = G/H$ is odd-dimensional and H is connected, then $\text{ind}_G P = 0$ for all elliptic P . The proof: the maximal torus $T \subset H$ gives an injection $i^* : K_H(V) \rightarrow K_T(V)$; since $\dim V = \dim M - \dim H / \dim H$ is odd, Bott periodicity gives $K_T(V) = 0$; hence $K_H(V) = 0$.
[¹]

Section 4: The Topological Index

The Goal: A Cohomological Formula for ind_a

The analytical index gives a homomorphism $\text{ind}_a : K_H(V) \rightarrow \mathbb{R}$. To compute it without solving any operator equations, one constructs the *topological index* $\text{ind}_t : K_H(V) \rightarrow \mathbb{R}$ using characteristic classes, and then proves (in Section 5) that $\text{ind}_t = \text{ind}_a$.

Relative Lie Algebra Cohomology

The relevant cohomology for homogeneous spaces G/H is not de Rham cohomology of M , but the *relative Lie algebra cohomology* $H^*(\mathfrak{g}, H; \mathbb{R})$ of the pair $(\mathfrak{g}, H)$. Its cochain complex consists of

$$C^q(\mathfrak{g}, H; \mathbb{R}) = \{\alpha \in \Lambda^q \mathfrak{g}^* : \iota_X \alpha = 0 \text{ for } X \in \mathfrak{h}, \text{ and } \text{Ad}^*(h)\alpha = \alpha \text{ for } h \in H\},$$

with differential $d\alpha(X_1, \dots, X_{q+1}) = \sum_{i < j} (-1)^{i+j} \alpha([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_{q+1})$. These are the H -invariant forms on $\mathfrak{g}/\mathfrak{h}$, and they correspond to G -invariant differential forms on $M = G/H$. [¹]

The Chern Character in $H^*(\mathfrak{g}, H; \mathbb{R})$

Fix an $\text{Ad}(H)$ -invariant splitting $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, giving a G -invariant connection on the principal bundle $H \rightarrow G \rightarrow M$ with connection form $\theta : \mathfrak{g} \rightarrow \mathfrak{h}$ (projection parallel to $\mathfrak{m}$) and curvature form $\Omega(X, Y) = -\frac{1}{2}[X, Y]_{\mathfrak{h}}$ for $X, Y \in \mathfrak{m}$.

For a unitary representation $\varepsilon : H \rightarrow U(E)$ with Lie algebra differential $\varepsilon : \mathfrak{h} \rightarrow \mathfrak{gl}(E)$, define the curvature form twisted by ε :

$$\Omega_\varepsilon(X, Y) = i\varepsilon(\Omega(X, Y)), \quad X, Y \in \mathfrak{m}.$$

The *Chern character* of ε in relative cohomology is

$$\text{ch}(\varepsilon) = [\text{tr exp}(\Omega_\varepsilon)] \in H^*(\mathfrak{g}, H; \mathbb{R}).$$

The form $\text{tr exp}(\Omega_\varepsilon)$ is H -invariant (since conjugating ε by h multiplies by a factor that cancels in the trace), pulled back to $\Lambda^* \mathfrak{g}^*$ via the projection $\mathfrak{g} \twoheadrightarrow \mathfrak{m}$, and is independent of the choice of splitting by the Chern-Weil argument. [¹]

The $\hat{A}$ -Class Analog

The analogue of the Hirzebruch $\hat{A}$ -polynomial in this setting is constructed from the co-isotropy representation V (the real H -module $\mathfrak{g}^*/\mathfrak{h}^0 \cong \mathfrak{m}^*$). One forms the curvature Ω_V for the complexification $V \otimes \mathbb{C}$, and defines

$$\hat{A}(\mathfrak{g}, H) = \left[\det^{1/2} \frac{\Omega_V/2}{\sinh(\Omega_V/2)} \right] \in H^*(\mathfrak{g}, H; \mathbb{R}).$$

(In the Dirac operator case, the precise form is $\hat{A}(\mathfrak{g}, H)$ itself; for the signature operator the analogous class is $\hat{E}(\mathfrak{g}, H)$ defined by $\det^{1/2}(\frac{2\Omega_V}{\tanh(\Omega_V/2)})$.) Since $\dim H^m(\mathfrak{g}, H; \mathbb{R}) = 1$ (as both G and H are unimodular), there is a canonical pairing: for a class $\Omega \in H^*(\mathfrak{g}, H; \mathbb{R})$, the scalar $\Omega[V]$ is defined by the relation $\Omega^{(m)} = (\Omega[V]) \cdot \omega_0$ where ω_0 is the volume element.[^1]

The K-Theory Structure and the Topological Index Formula

The Bott class construction (Lemma 4.1, Proposition 4.3) shows: if G/H admits a G -invariant spin structure (i.e., $\text{Ad}^* : H \rightarrow SO(V)$ lifts to $\text{Spin}(V)$), then $K_H(V) \cong R(H)$ as an $R(H)$ -module, freely generated by the Dirac Bott class β . Every $k \in K_H(V)$ can be written uniquely as $k = a\beta$ for some $a \in R(H)$. When there is no spin structure, one passes to the canonical double covering $\tilde{H} \subset H \times \text{Spin}(V)$, obtains a $\mathbb{Z}_2$ -graded module structure, and still decomposes every $K_H(V)$ -class as $a\beta$ for $a \in R(\tilde{H})^1$.[^1]

The *topological index* is then defined: for $k = a\beta \in K_H(V)$,

$$\boxed{\text{ind}_t k := (\text{ch}(a) \cdot \hat{A}(\mathfrak{g}, H))[V].}$$

The characteristic class $\text{ch}(a) \cdot \hat{A}(\mathfrak{g}, H)$ lives in $H^*(\mathfrak{g}, H; \mathbb{R})$, and $(-)[V]$ evaluates the top-degree component against the volume. The formula is independent of the choices of splitting and volume element (up to the coherence condition $dg/dh = \omega_{\text{vol}}$ fixed in Section 5).[^1]

Section 5: The Index Theorem — Heat Kernel Proof

Main Statement

Theorem 5.3 (The L^2 -Index Theorem). *For any unimodular Lie group G , compact subgroup H , and G -invariant elliptic ΨDO $P \in \Psi_c^n(M, \xi_1, \xi_2)^G$,*

$$\text{ind}_G P = \text{ind}_t[\sigma(P)].$$

The proof is carried out in two stages: first for signature operators twisted by homogeneous bundles (Theorem 5.1), then for Dirac-type operators (Theorem 5.2), from which the general case follows by the K-theory decomposition of Section 4.[^1]

Stage 1: The Signature Operator (Theorem 5.1)

Fix a G -invariant Riemannian metric on M and let $D = d + d^*$ split into two operators $D^\pm : C_c^\infty(M, \Lambda^\pm T_{\mathbb{C}}^* M) \rightarrow C_c^\infty(M, \Lambda^\mp T_{\mathbb{C}}^* M)$ via the Hodge star involution. For a finite-dimensional unitary H -representation ε with bundle ξ , form the twisted operators $D_\varepsilon^\pm$ with $\Delta^\pm = (D_\varepsilon^\pm)^* D_\varepsilon^\pm$. The theorem claims

$$\text{ind}_G D_\varepsilon^+ = 2(\text{ch}(\varepsilon) \cdot \hat{E}(\mathfrak{g}, H))[V].$$

Step 1: The heat operator $e^{-t\Delta^+}$ is of G-trace class. Because Δ^+ is a G -invariant elliptic operator of order 2, the heat kernel $e^{-t\Delta^+}$ is a smoothing operator (it has a smooth kernel for each $t > 0$). By Lemma 2.2, any such compactly-based smoothing operator is in $\text{Dom}(\text{Av}_G)$, and by Lemma 2.1, $\text{tr}_G(e^{-t\Delta^+}) < \infty$. [1]

Step 2: Asymptotic expansion of the G-trace. The core claim is

$$\text{tr}_G e^{-t\Delta^+} \sim \sum_{k \geq -m} c_k(\Delta^+) t^{k/2} \quad \text{as } t \rightarrow 0^+,$$

whose coefficients are local invariants of the operator (computable from the operator's coefficients in any coordinate chart). This is proved by comparing $\text{tr}_G e^{-t\Delta^+}$ to the localized ordinary trace $\text{tr}(E(t))$, where $E(t) = \frac{1}{2\pi i} \int_\Gamma e^{-t\lambda} P_\lambda d\lambda$ is the contour integral expression for the heat semigroup via the resolvent P_λ of Δ^+ . The comparison (equation 5.1.7) gives $|\text{tr}_G e^{-t\Delta^+} - \text{tr}(E(t))| = O(t^k)$ for any k , reducing the problem to the well-known classical asymptotic expansion of $\text{tr}(E(t))$ [1][11].

Step 3: The McKean-Singer identity with G-trace. The key algebraic identity is

$$\text{tr}_G e^{-t\Delta^+} - \text{tr}_G e^{-t\Delta^-} = \dim_G \ker D_\varepsilon^+ - \dim_G \ker D_\varepsilon^- = \text{ind}_G D_\varepsilon^+$$

for all $t > 0$. This is the G-trace version of the McKean-Singer formula. The proof is as in the classical case: $\text{tr}_G e^{-t\Delta^+} - \dim_G \ker D^+ = \text{tr}_G((I - H^+)e^{-t\Delta^+}(I - H^+))$, where $H^\pm$ is the projection onto $\ker D^\pm$. Using the polar decomposition $D^\pm = U^\pm |D^\pm|$, one shows the off-diagonal terms cancel, yielding the identity. The critical point is that this identity holds for all $t > 0$, so it is t -independent. In particular, taking $t \rightarrow 0$, the left-hand side equals the constant term in the asymptotic expansion of the G-trace difference [1].

Step 4: Local computation of the constant term. The constant term in $\text{tr}_G e^{-t\Delta^+} - \text{tr}_G e^{-t\Delta^-}$ is

$$c_0(\Delta^+) - c_0(\Delta^-) = \int_M \Omega(U),$$

where $\Omega(U)$ is the restriction to a coordinate chart U of the global G -invariant form

$$\Omega = 2^n \left(\text{ch}(\varepsilon) \cdot \hat{E}(\mathfrak{g}, H) \right) \cdot \omega_{\text{vol}}.$$

This follows from the local version of the generalized signature theorem applied to the G -invariant curvature forms Ω_ε and Ω_V . Integrating over M and using the normalization $\int_G f dg = 1$ for the cut-off function yields the stated formula. [1]

Stage 2: The Dirac Operator (Theorem 5.2)

For representations ε with $[\varepsilon] \in R(\tilde{H})^1$ (i.e., the spinor side in the $\mathbb{Z}_2$ -graded structure), one forms the $(\pm)$ -Dirac operators $A_\varepsilon^\pm$ using Clifford multiplication. Theorem 5.2 states

$$\mathrm{ind}_G A_\varepsilon^+ = (-1)^n (\mathrm{ch}(\varepsilon) \cdot \hat{A}(\mathfrak{g}, H)) [V].$$

The proof proceeds in three steps:

Step 2a (Special case $\varepsilon = \mu \otimes \sigma$): When $\varepsilon = \mu \otimes \sigma$ where $\mu \in R(H)^0$ and $\sigma = \sigma^+ \oplus \sigma^-$ is the full spin representation, one shows the principal symbols of $A_{\mu, \sigma}^+$ and D_μ^+ (the twisted signature operator) agree up to the factor $(-1)^n$, hence their G-indices coincide up to sign. Applying Theorem 5.1 gives the formula in this special case.[^1]

Step 2b (Polynomial dependence and I_H -adic continuity): One shows the map $\delta_\varepsilon : R(\tilde{H})^1 \rightarrow \mathbb{R}$, $\varepsilon \mapsto \mathrm{ind}_G A_\varepsilon^+$, is a polynomial function of the matrix coefficients of ε . Explicitly: by computing the local expression of A_ε^+ in exponential coordinates (equation 5.2.2), the local heat kernel measures $\mu^\pm(U)$ are polynomial in $\varepsilon(Y_1), \dots, \varepsilon(Y_r)$ for a basis Y_j of $\mathfrak{h}$. The index is then a polynomial in these matrix coefficients (equation 5.2.3).[^1]

The Weyl invariant theory for the unitary group then implies that any polynomial invariant under $U(E)$ is a linear combination of products of traces $\mathrm{tr}(\varepsilon(Y_{i_1}) \cdots \varepsilon(Y_{i_s}))$. Since these depend only on $(\chi(\varepsilon) - \dim \varepsilon)^p$ -type combinations (i.e., on the augmentation-ideal filtration in $R(H)$), one deduces that δ_ε is continuous in the $I_{\tilde{H}}$ -adic topology — it vanishes on high powers of the augmentation ideal.[^1]

Step 2c (Convergence argument): The element $\sigma \otimes \sigma = A_{\mathbb{C}}^+ V_c$ equals $2^m [\chi]$ where χ is the representation on $\Lambda^* V_{\mathbb{C}}$. Since $2^m [\chi]^{-1}$ is the limit of a sequence $\{C_k\}$ in $(R(\tilde{H})^0 \otimes \mathbb{Q})^\wedge$ (the augmentation-adic completion), and since the formula in Step 2a together with I_H -adic continuity allows passage to the limit, one obtains the formula for all $\varepsilon \in R(\tilde{H})^1$.[^1]

Stage 3: General Elliptic Operators (Theorem 5.3)

By Proposition 4.3, every class $k \in K_H(V)$ is uniquely of the form $k = a\beta$ with $a \in R(\tilde{H})^1$. So every elliptic symbol class can be matched to a Dirac operator with the same symbol class in $K_H(V)$. By Proposition 3.5 (the analytical index depends only on the K-theory symbol class) and Theorem 5.2 (the Dirac operator result), the formula $\mathrm{ind}_G P = \mathrm{ind}_t[\sigma(P)]$ holds for all elliptic G -invariant Ψ DOs.[^1]

Section 6: Decomposition of the Kernel into Discrete Series

The Second Main Result

Theorem 6.1. *Let G be a semisimple Lie group with finite center, $H \subset G$ a compact subgroup, and $P \in \Psi_c^n(M, \xi_1, \xi_2)^G$ elliptic. Then $\ker P$ decomposes as a finite direct sum of irreducible discrete series representations of G .*

A *discrete series representation* of G is an irreducible unitary representation that appears as a direct summand in $L^2(G)$ (equivalently, its matrix coefficients are in $L^2(G)$). They have a *formal degree* $\deg(\pi) > 0$, which is the "dimension per unit volume" of the representation space.[6][1]

Proof Strategy

The proof uses the Plancherel decomposition of $L^2(G)$: any $f \in L^2(G)$ can be expanded as

$$f = \int_{\hat{G}} \hat{f}(\pi) d\mu(\pi),$$

where $\hat{G}$ is the unitary dual of G , μ is the Plancherel measure, and $\hat{f}(\pi)$ is the operator-valued Fourier transform. The representation space $\ker P$ is a closed G -invariant subspace, so it decomposes as a direct integral

$$\ker P \cong \int_{\hat{G}} \mathcal{C}(\pi) \otimes \ker(I - T_\pi(\Delta)) d\mu(\pi),$$

where T_π is the action of the parametrix remainder on the π -isotypic component.

The key step is showing the continuous part of the Plancherel measure contributes nothing. For non-discrete-series representations π , Harish-Chandra's detailed analysis of the Plancherel measure shows that the operator $I - T_\pi(\Delta)$ is invertible for μ -almost all such π . This uses the *G-invariant parametrix*: the existence of a G -invariant parametrix Q such that $I - QP$ is G -invariant and smoothing means that for the Fourier transform, $I - T_\pi(QP) = \hat{S}_1(\pi)$ is Hilbert-Schmidt for all π . For the continuous spectrum, $\|\hat{S}_1(\pi)\|_{HS} > 1$ would lead (via the Plancherel formula $\mathrm{tr}_G(S_1^* S_1) = \int \|T_\pi(S_1)\|_{HS}^2 d\mu(\pi)$) to a contradiction with $\dim_G \ker P < \infty$. Therefore the continuous part contributes zero [^1].

For the discrete part, the G-index formula gives $\sum_\pi \dim(\ker P|_\pi) \deg(\pi) \leq \dim_G \ker P < \infty$. Since formal degrees are bounded away from zero (they are positive for discrete series), the sum can have only finitely many nonzero terms [^1].

Theorem 6.2 extends this result from semisimple G to a larger class: G is a semi-direct product of an " $\mathcal{N}$ -group" (a solvable group with properties similar to nilpotent groups with discrete series) with a connected reductive Lie group S , under suitable conditions on the compact subgroup H . [^1]

The Vanishing Criterion

Corollary 6.3. *Under the hypotheses of Theorem 6.2, $\ker P = \{0\}$ if $\mathrm{rank}(K_S) < \mathrm{rank}(S)$, where K_S is a maximal compact subgroup of the reductive factor S .*

For semisimple groups, Harish-Chandra proved that G has discrete series if and only if it has a compact Cartan subgroup, which is equivalent to $\mathrm{rank}(K) = \mathrm{rank}(G)$. The corollary says: when this rank condition fails, the discrete series is empty, so no elliptic equation can have nontrivial L^2 -solutions. [^1]

Section 7: Applications

7.1 — Nilpotent Groups: All Discrete Series via L^2 -Harmonic Spinors

Let G be a connected, simply-connected nilpotent Lie group with center Z , and assume G has square-integrable (mod Z) irreducible representations. The Kirillov correspondence parametrizes these by regular coadjoint orbits: a linear functional $\lambda \in \mathfrak{g}^*$ determines a character e_λ of Z and an irreducible representation π_λ that is square-integrable iff the isotropy group $G(\lambda)$ equals $\mathfrak{z}$ (the center).

For each regular λ , form the homogeneous line bundle ξ_λ over $M = G/Z$ (the unique spin structure on this even-dimensional nilpotent homogeneous space), and construct the Dirac operators $A_\lambda^\pm$. Proposition 7.1A shows:

$$m^+ - m^- = 1,$$

where $m^\pm =$ multiplicity of π_λ in $\ker A_\lambda^\pm$. The proof uses the index formula:

$$\text{ind}_G A_\lambda^+ = \frac{1}{n! \cdot 2^n} (\Lambda^n \beta_\lambda)[V], \quad n = \frac{1}{2} \dim M,$$

and the Kirillov formula $\deg(\pi_\lambda) = \frac{1}{n! \cdot 2^n} (\Lambda^n \beta_\lambda)[V]$, giving $\text{ind}_G A_\lambda^+ = \deg(\pi_\lambda)$ and hence $(m^+ - m^-) \deg(\pi_\lambda) = \deg(\pi_\lambda)$, so $m^+ - m^- = 1$.^[1]

Proposition 7.1B further shows that for sufficiently large integer character (large p), $\ker A^-$ vanishes entirely, so $\ker A^+$ is exactly the representation space of π_λ . The proof uses a Bochner-type inequality:

$$(A_p^-)^* A_p^- \geq (2\pi p - n^2 2^{n-2} C^2) I,$$

which becomes positive for p large enough, forcing $\ker A_p^- = 0$.^[1]

7.2 — Non-Integrality of the G-Index

The paper provides an explicit counterexample showing the G-index need not be an integer for general G . The 4-dimensional nilpotent group G with Lie algebra $\mathfrak{g} = \text{span}\{x, y, z, w\}$ and bracket $[x, y] = z + \theta w$ (θ irrational) has discrete series parametrized by pairs $(p, q) \in \mathbb{Z}^2 \setminus \{0\}$ with formal degrees $\deg(\pi_{p,q}) = |p + \theta q|$. The corresponding Dirac operator satisfies $\text{ind}_G A_{p,q}^+ = |p + \theta q|$, which is not bounded away from zero and is generically irrational. Hence no normalization of Haar measure can make all G-indices integers simultaneously.^[1]

7.3 — Reciprocity Theorem

For a G -invariant elliptic differential operator $D : C^\infty(M, \xi_1) \rightarrow C^\infty(M, \xi_2)$ and a unitary representation π of G , define the "representation-twisted" operator $D_\pi : (JC^\infty(\pi) \otimes E_1)^H \rightarrow (JC^\infty(\pi) \otimes E_2)^H$ (acting on smooth vectors of π tensored with the bundle fibres, restricted to H -fixed vectors). Proposition 7.3A proves:

- (i) $\ker D_\pi = 0$ for Plancherel-almost all non-discrete-series π ;
- (ii) $\ker D_\pi = 0$ for all but finitely many discrete series π ;

(iii) The G-index satisfies:

$$\mathrm{ind}_G D = \sum_{\pi \in G_d} (\dim \ker D_\pi - \dim \ker D_\pi^*) \deg(\pi).$$

This is a *reciprocity formula*: the global G-index breaks into a sum over discrete series weighted by formal degrees. For semisimple G with maximal compact K (Corollary 7.3B), it specializes to

$$\mathrm{ind}_G D = \sum_{\pi} ([\pi : \varepsilon_1] - [\pi : \varepsilon_2]) \deg(\pi),$$

where $[\pi : \varepsilon_i]$ is the multiplicity of ε_i in $\pi|_K [^\wedge 1]$.

The Logical Architecture of the Proof

The seven sections interlock as follows. Section 1 builds the Ψ DO calculus needed to define all objects. Section 2 establishes the G-trace, the central measurement device, and relates it to local data via the key identity $\mathrm{tr}_G \circ \mathbf{Av}_G = \mathrm{tr}$. Section 3 defines the analytical index, proves it is finite-valued (the G-dimension of the kernel), and shows it factors through K-theory. Section 4 constructs the topological index via Chern-Weil theory in relative Lie algebra cohomology and the Bott-class decomposition of $K_H(V)$. Section 5 proves equality of the two indexes by the heat kernel method: the McKean-Singer identity converts the analytical index into the constant term of the heat-trace asymptotics, while the local calculation converts that constant term into the topological index formula. Sections 6 and 7 exploit the machinery for representation-theoretic consequences.^{[10][1]}

The most subtle technical point is in Section 5, Step 2b: the index depends on the symbol polynomially in matrix coefficients, but the Bott class decomposition requires handling representations not in $R(H)^0 \otimes \mathbb{Q}$. Weyl's fundamental theorem of invariant theory for the unitary group reduces all polynomial invariants to traces, and I_H -adic continuity (the representation-ring analogue of Taylor approximation by characters) then allows the index formula to be extended from the special class of representations $\mu \otimes \sigma$ to all of $R(\tilde{H})^1 [^\wedge 1]$.

Place in Mathematics

This paper occupies a precise position in the genealogy of index theory: it is the synthesis of Atiyah-Singer (compact case), Atiyah's covering L^2 -index (discrete groups), and Harish-Chandra's representation theory (discrete series construction), extended to the full non-compact homogeneous setting without any lattice assumption. It was also, historically, a model for Connes's subsequent noncommutative geometry program: the abstract structure (G-trace on a von Neumann algebra, factorization of the index through K-theory, characteristic class formula) prefigures the local index formula in noncommutative geometry (1995), where the same ingredients appear in the spectral triple framework for arbitrary noncommutative spaces. The 2024 paper of H. Wang and Z. Wang further generalizes to almost-connected Lie groups and proper cocompact actions using the same heat kernel machinery.^{[12][13][^\wedge 10]}

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